Digital Electronics - Previous Year Paper (Winter 2023)

College Buddy - free study resource sample file

Q1. Convert (1011.101)2 into decimal and octal. Show all steps.
Q2. Simplify the Boolean function using a 4-variable K-map.
Q3. Design a full adder using two half adders. Draw the circuit.
Q4. Explain the working of a JK flip-flop with its truth table.
Q5. Design a 3-bit synchronous counter and draw the timing diagram.
Q6. Differentiate between combinational and sequential circuits.